Equivalent Fractions and Mixed Numbers

Name _____ Date _____ No calculator Version 1

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KEY WORDS numerator *(taurunga)* · denominator *(tauraro)* · equivalent fraction *(hautanga ōrite)*

A Equivalent fractions ●○○

Remember: Multiply or divide the top and the bottom by the same number: $\frac{2}{3} = \frac{2 \times 4}{3 \times 4} = \frac{8}{12}$.

1. Fill in the box to make the fractions equal.

a) $\frac{1}{2} = \frac{\square}{8}$ **b)** $\frac{2}{3} = \frac{\square}{9}$ **c)** $\frac{3}{4} = \frac{\square}{20}$ **d)** $\frac{2}{5} = \frac{8}{\square}$

e) $\frac{5}{6} = \frac{\square}{12}$ **f)** $\frac{3}{8} = \frac{9}{\square}$ **g)** $\frac{7}{10} = \frac{\square}{100}$ **h)** $\frac{4}{9} = \frac{16}{\square}$

i) $\frac{1}{3} = \frac{\square}{15}$ **j)** $\frac{3}{5} = \frac{18}{\square}$ **k)** $\frac{5}{8} = \frac{\square}{24}$ **l)** $\frac{2}{7} = \frac{10}{\square}$

B Simplifying fractions ●○○

Watch out: Divide the top and the bottom by the same number, and keep going until no number divides into both.

2. Write each fraction in its simplest form.

a) $\frac{4}{8}$ = _____ **b)** $\frac{6}{9}$ = _____ **c)** $\frac{10}{15}$ = _____ **d)** $\frac{12}{16}$ = _____

e) $\frac{8}{20}$ = _____ **f)** $\frac{15}{25}$ = _____ **g)** $\frac{18}{24}$ = _____ **h)** $\frac{14}{21}$ = _____

i) $\frac{30}{40}$ = _____ **j)** $\frac{24}{36}$ = _____ **k)** $\frac{9}{12}$ = _____ **l)** $\frac{28}{42}$ = _____

C Improper fractions and mixed numbers ●●○

Remember: An improper fraction has a numerator larger than its denominator. $\frac{7}{4}$ is 7 quarters, which makes 1 whole and 3 quarters: $1\frac{3}{4}$.

3. Write each improper fraction as a mixed number.

a) $\frac{7}{4}$ = _____ **b)** $\frac{5}{2}$ = _____ **c)** $\frac{11}{3}$ = _____ **d)** $\frac{9}{5}$ = _____

e) $\frac{17}{6}$ = _____ **f)** $\frac{23}{10}$ = _____ **g)** $\frac{13}{4}$ = _____ **h)** $\frac{19}{8}$ = _____

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4. Write each mixed number as an improper fraction.

a) $1\frac{1}{2}$ = _____ **b)** $2\frac{1}{3}$ = _____ **c)** $1\frac{3}{4}$ = _____ **d)** $3\frac{2}{5}$ = _____

e) $2\frac{5}{6}$ = _____ **f)** $4\frac{1}{2}$ = _____ **g)** $1\frac{7}{10}$ = _____ **h)** $3\frac{3}{8}$ = _____

D Comparing fractions ●●●

Remember: Change the fractions so that they have the same denominator, then compare the numerators.

5. Which fraction is larger?

a) $\frac{1}{2}$ or $\frac{3}{8}$ **b)** $\frac{2}{3}$ or $\frac{5}{6}$ **c)** $\frac{3}{4}$ or $\frac{5}{8}$

d) $\frac{2}{5}$ or $\frac{3}{10}$ **e)** $\frac{1}{3}$ or $\frac{2}{5}$ **f)** $\frac{3}{4}$ or $\frac{2}{3}$

6. Write each set of fractions in order, smallest first.

a) $\frac{1}{2}$, $\frac{3}{8}$, $\frac{1}{4}$ **b)** $\frac{2}{3}$, $\frac{5}{6}$, $\frac{1}{2}$ **c)** $\frac{3}{5}$, $\frac{1}{2}$, $\frac{7}{10}$

Extension

7. Write three different fractions that are equivalent to $\frac{3}{4}$.

8. Which is larger, $\frac{5}{8}$ or $\frac{2}{3}$? Show your working.

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Equivalent Fractions and Mixed Numbers

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

KEY WORDS numerator *(taurunga)* · denominator *(tauraro)* · equivalent fraction *(hautanga ōrite)*

A Equivalent fractions ●○○

Remember: Multiply or divide the top and the bottom by the same number: $\frac{2}{3} = \frac{2 \times 4}{3 \times 4} = \frac{8}{12}$.

1. Fill in the box to make the fractions equal.

a) $\frac{1}{3} = \frac{\square}{12}$ **b)** $\frac{3}{4} = \frac{\square}{12}$ **c)** $\frac{2}{5} = \frac{\square}{25}$ **d)** $\frac{3}{5} = \frac{12}{\square}$

e) $\frac{5}{8} = \frac{\square}{16}$ **f)** $\frac{4}{7} = \frac{12}{\square}$ **g)** $\frac{9}{10} = \frac{\square}{100}$ **h)** $\frac{5}{9} = \frac{20}{\square}$

i) $\frac{1}{2} = \frac{\square}{14}$ **j)** $\frac{2}{3} = \frac{12}{\square}$ **k)** $\frac{3}{8} = \frac{\square}{40}$ **l)** $\frac{4}{5} = \frac{20}{\square}$

B Simplifying fractions ●○○

Watch out: Divide the top and the bottom by the same number, and keep going until no number divides into both.

2. Write each fraction in its simplest form.

a) $\frac{3}{6}$ = _____ **b)** $\frac{8}{12}$ = _____ **c)** $\frac{10}{25}$ = _____ **d)** $\frac{12}{18}$ = _____

e) $\frac{6}{20}$ = _____ **f)** $\frac{20}{25}$ = _____ **g)** $\frac{16}{24}$ = _____ **h)** $\frac{21}{28}$ = _____

i) $\frac{30}{50}$ = _____ **j)** $\frac{27}{36}$ = _____ **k)** $\frac{6}{9}$ = _____ **l)** $\frac{32}{48}$ = _____

C Improper fractions and mixed numbers ●●○

Remember: An improper fraction has a numerator larger than its denominator. $\frac{7}{4}$ is 7 quarters, which makes 1 whole and 3 quarters: $1\frac{3}{4}$.

3. Write each improper fraction as a mixed number.

a) $\frac{9}{4}$ = _____ **b)** $\frac{7}{2}$ = _____ **c)** $\frac{13}{3}$ = _____ **d)** $\frac{8}{5}$ = _____

e) $\frac{19}{6}$ = _____ **f)** $\frac{27}{10}$ = _____ **g)** $\frac{15}{4}$ = _____ **h)** $\frac{21}{8}$ = _____

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4. Write each mixed number as an improper fraction.

a) $1\frac{1}{3}$ = _____ **b)** $2\frac{1}{2}$ = _____ **c)** $1\frac{2}{5}$ = _____ **d)** $3\frac{3}{4}$ = _____

e) $2\frac{1}{6}$ = _____ **f)** $5\frac{1}{2}$ = _____ **g)** $1\frac{9}{10}$ = _____ **h)** $2\frac{5}{8}$ = _____

D Comparing fractions ●●●

Remember: Change the fractions so that they have the same denominator, then compare the numerators.

5. Which fraction is larger?

a) $\frac{1}{2}$ or $\frac{5}{8}$ **b)** $\frac{1}{3}$ or $\frac{1}{6}$ **c)** $\frac{1}{4}$ or $\frac{3}{8}$

d) $\frac{3}{5}$ or $\frac{7}{10}$ **e)** $\frac{2}{3}$ or $\frac{3}{5}$ **f)** $\frac{3}{4}$ or $\frac{4}{5}$

6. Write each set of fractions in order, smallest first.

a) $\frac{1}{2}$, $\frac{5}{8}$, $\frac{3}{4}$ **b)** $\frac{1}{3}$, $\frac{1}{6}$, $\frac{1}{2}$ **c)** $\frac{2}{5}$, $\frac{1}{2}$, $\frac{3}{10}$

★ Extension

7. Write three different fractions that are equivalent to $\frac{2}{3}$.

8. Which is larger, $\frac{5}{6}$ or $\frac{7}{9}$? Show your working.

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